
Geophysics Inverse Problem

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Assignment: Number 1

Problem 1 Prove the formula of the projection operator $\mathbf{P}_y \mathbf{x} = \frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}}$, where $\mathbf{P}_y$ denotes the orthogonal projection onto $\text{span}(\mathbf{y})$.

(a) Let $\mathcal{U} = \text{span}(\mathbf{y})$. By orthogonal decomposition, $\mathbf{x}$ can be written as

$$\alpha \mathbf{y} + (\mathbf{x} - \alpha \mathbf{y}),$$

where, we choose α such that

$$(\mathbf{x} - \alpha \mathbf{y}) \in \mathcal{U}^\perp$$

(b) The orthogonality condition gives

$$(\mathbf{x} - \alpha \mathbf{y})^\top \mathbf{y} = 0.$$

Thus

$$\alpha = \frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}}.$$

(c) Therefore,

$$\mathbf{x} = \frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \mathbf{y} + (\mathbf{x} - \frac{\mathbf{x}^\top \mathbf{y}}{\mathbf{y}^\top \mathbf{y}} \mathbf{y})$$

The first term is the orthogonal projection of $\mathbf{x}$ onto $\text{span}(\mathbf{y})$, and the second term is orthogonal to $\mathbf{y}$

Problem 2 Derive the identities $\frac{\partial(\mathbf{x}^\top \mathbf{A})}{\partial \mathbf{x}} = \mathbf{A}^\top$ and $\nabla \mathbf{x}^\top \mathbf{A} = \mathbf{A}$.

$$\begin{aligned} & \frac{\partial(\mathbf{x}^\top \mathbf{A})}{\partial \mathbf{x}} \\ &= \frac{\partial x_j \mathbf{e}_j \cdot \mathbf{A}_{kl} \mathbf{e}_k \mathbf{e}_l}{\partial x_i} \mathbf{e}_i \\ &= \delta_{ij} \mathbf{A}_{kl} \delta_{jk} \mathbf{e}_l \mathbf{e}_i \\ &= \delta_{jk} \mathbf{A}_{kl} \mathbf{e}_l \mathbf{e}_j \\ &= \mathbf{A}_{jl} \mathbf{e}_l \mathbf{e}_j \\ &= \mathbf{A}^\top \end{aligned}$$

$$\begin{aligned} & \nabla \mathbf{x}^\top \mathbf{A} \\ &= \frac{\partial}{\partial x_i} \mathbf{e}_i x_j \mathbf{e}_j \cdot \mathbf{A}_{kl} \mathbf{e}_k \mathbf{e}_l \\ &= \mathbf{e}_i \delta_{ij} \mathbf{A}_{kl} \delta_{jk} \mathbf{e}_l \\ &= \mathbf{A}_{jl} \delta_{ij} \mathbf{e}_l \mathbf{e}_i \\ &= \mathbf{A}_{il} \mathbf{e}_i \mathbf{e}_l \\ &= \mathbf{A} \end{aligned}$$